@ codedsoul_05

DA NOTES v1.0 ★

DATA ANALYSIS

BEGINNER → ADVANCED

COMPLETE PREMIUM NOTES

From raw data to clear insights — clean, explore, visualize, and decide with confidence.



① COLLECT → ② CLEAN → ③ EXPLORE → ④ DECIDE

"Ask better questions. Trust the numbers."

Stats • SQL • Excel • Python • Viz • EDA • Storytelling

① WHAT'S INSIDE

- Fundamentals
- Data Types
- Cleaning
- EDA
- Stats Basics
- SQL for Analysis
- Excel/Sheets
- Python (pandas)
- Visualization
- Metrics & KPIs
- Experiments
- Recap

② BY THE NUMBERS

- 40+ Topics
- 50+ Charts Ideas
- Interview Frameworks
- Beginner to Pro
- Interview Ready
- Real Workflows

③ TOPICS COVERED

- ☑ Data Cleaning
- ☑ Descriptive Stats
- ☑ Distributions
- ☑ Correlation
- ☑ SQL Aggregations
- ☑ pandas EDA
- ☑ Charts that work
- ☑ Dashboards mindset
- ☑ Bias & pitfalls
- ☑ Storytelling with data

★ Clean data. Clear charts. Honest insights. ★

01 — WHAT & WHY DATA ANALYSIS

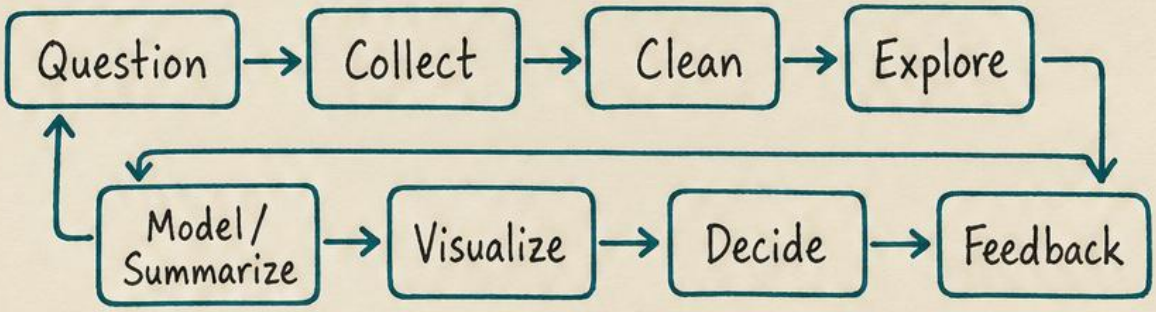
GOAL

Turn raw data into decisions you can defend.

THE PROBLEM

Gut feel, messy sheets, wrong charts, biased conclusions.

ANALYSIS LOOP



WHO USES IT

Product, Marketing, Finance, Ops, Research.

TIP

Start with a decision, not a dataset.

★ Insight without action is just trivia. ★

02 — DATA TYPES & STRUCTURES

QUANTITATIVE

- Numerical data
- Arithmetic makes sense

Examples: age, height, income, temperature

VS

QUALITATIVE

- Categorical data
- Labels or groups

Examples: gender, color, product type, country

DISCRETE

- Countable values
- Usually whole numbers

Examples: number of students, orders, defects

CONTINUOUS

- Measurable on a continuum
- Can take any value

Examples: height, weight, time, distance

LEVELS OF MEASUREMENT (SCALES)

NOMINAL

Categories with no order

Examples: color, country

ORDINAL

1 —
2 — Categories with
3 — a meaningful order

Examples: rank, satisfaction level

INTERVAL

−2 −1 0 1 2

Ordered on equal intervals, no true zero

Examples: °C, °F, calendar year

RATIO

Ordered, equal intervals, true zero

Examples: height, weight, income

TABULAR DATA ANATOMY

- Rows = observations (records)
- Columns = variables (attributes)

ID	Name	Age	City
1	Riya	24	Delhi
2	Arjun	28	Pune
3	Meera	22	Bhopal
...	...	...	...

ROWS (observations)

— COLUMNS (variables)

WIDE vs LONG FORMAT

WIDE FORMAT

ID	Name	Math	Sci	Eng
1	Riya	85	90	88
2	Arjun	78	82	80
3	Meera	92	95	93
...	...	...	...	...

→ Better for humans to read

LONG FORMAT

ID	Name	Subject	Score
1	Riya	Math	85
1	Riya	Sci	90
1	Riya	Eng	88
...	...	...	...

→ Better for analysis & modeling

MISSING VALUES: TYPES

- **MCAR** (Missing Completely At Random)
  Missingness is completely random.
  Example: data entry error.
- **MAR** (Missing At Random)
  Missingness related to observed data.
  Example: older people less likely to answer income.
- **MNAR** (Missing Not At Random)
  Missingness related to unobserved data itself.
  Example: people with very high income skip the question.

Design your data collection & cleaning with the missingness in mind.

★ KNOW YOUR VARIABLE TYPE BEFORE YOU PICK A CHART. ★

03 — DATA CLEANING

- [x] Fix types
- [x] Outliers (investigate first)
- [x] Trim text
- [x] Standardize categories
- [x] Handle nulls
- [x] Parse dates
- [x] Remove duplicates
- [x] Validate ranges

BEFORE → **AFTER**

ID	NAME	JOINED	AMOUNT
1	' alice '	2024/1/5	100
2	BOB	05-02-24	200.00
2	Bob	2024-02-05	200
3	Charlie	Feb 7, 2024	' 300
4	' david '	2024/02/30	-50
5	Eve	.	NaN
6	ALICE	2024-01-05	100

ID	NAME	JOINED	AMOUNT
1	Alice	2024-01-05	100.00
2	Bob	2024-02-05	200.00
3	Charlie	2024-02-07	300.00
4	David	2024-02-28	-50.00
5	Eve	NULL	NULL
6	Alice	2024-01-05	100.00

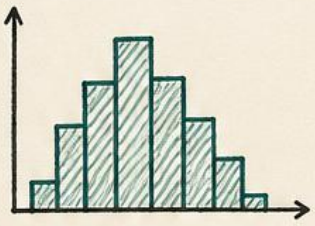
TIP: Document every cleaning step. Reproducible > clever.

Clean once. Trust forever.

04 — EDA WORKFLOW

① UNIVARIATE

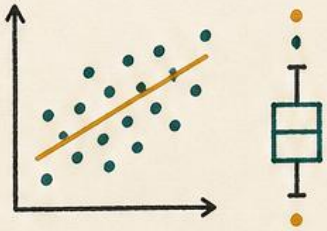
Understand each variable on its own.



- Distributions
- Summary stats
- Outliers
- Data types

② BIVARIATE

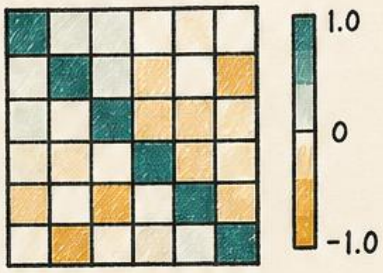
Explore relationships between pairs.



- Scatter / Line
- Boxplots
- Group comparisons
- Crosstabs

③ MULTIVARIATE

Understand patterns across many variables.



- Correlation heatmap
- Dimensionality reduction (PCA / t-SNE)
- Clustering

QUESTIONS TO ASK

- What's typical?
- What's weird?
- What relationships?
- What segments?

RED FLAGS

- **Selection bias** — Non-representative data leads to misleading insights.
- **Leakage** — Using future info (directly or indirectly) inflates performance.
- **Tiny samples** — Small n = noisy estimates, unstable conclusions.
- **Cherry-picked windows** — Handpicking time ranges can hide the real story.

Explore before you model. Curiosity first.

05 — DESCRIPTIVE STATS

MEAN MEDIAN MODE — WHEN TO USE EACH

- **MEAN** use when data is symmetric, no extreme outliers.
- **MEDIAN** use when data is skewed or has outliers.
- **MODE** use for categorical data or to find the most frequent value.

RANGE

Spread between the maximum and minimum.

$$\text{Range} = \text{Max} - \text{Min}$$

IQR ($Q_3 - Q_1$)

Spread of the middle 50% of the data.



$$\text{IQR} = Q_3 - Q_1$$

VARIANCE (σ^2)

Measures average squared deviation from the mean.

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

STD DEV (σ)

Square root of variance. In original units.

$$\sigma = \sqrt{\sigma^2}$$

PERCENTILES

Values below which a given % of observations fall.

Examples:

- 25th percentile = Q_1
- 50th percentile = Median
- 75th percentile = Q_3



SKEWNESS (BRIEF)

Direction and degree of asymmetry in data.

- Skew > 0 : right-skewed (long tail on right)
- Skew < 0 : left-skewed (long tail on left)
- Skew ≈ 0 : symmetric

OUTLIERS vs EXTREMES

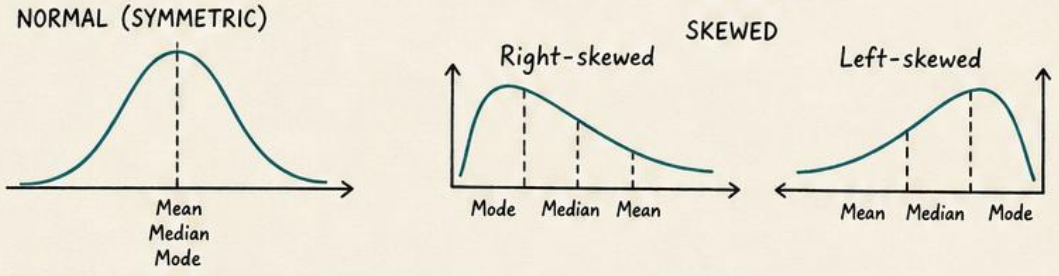
OUTLIERS

- Unusual points that stand out from the pattern.
- May be errors or natural variation.
- Detected using rules like 1.5 × IQR, Z-score, etc.

EXTREMES

- Very far values.
- Always rare, but not always outliers.
- All outliers are extremes, but not all extremes are outliers.

NORMAL vs SKEWED DISTRIBUTION



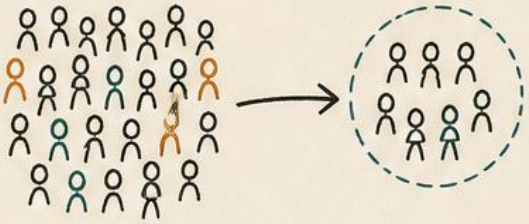
★ Median resists outliers. Mean does not. ★

06 — DISTRIBUTIONS & PROBABILITY

POPULATION vs SAMPLE

POPULATION
The whole group

SAMPLE
A subset

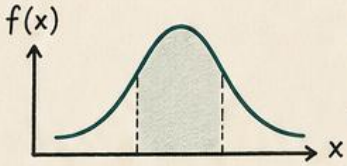


Parameter (μ, σ)

Statistic $(\bar{x}, \bar{s})$

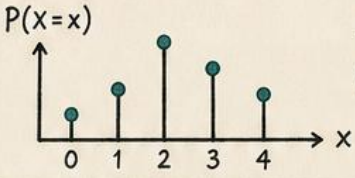
PDF / PMF — THE IDEA

PDF (continuous)



Area under the curve = probability

PMF (discrete)

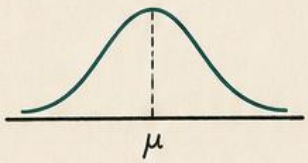


Sum of all probabilities = 1

COMMON DISTRIBUTIONS — AT A GLANCE

NORMAL (μ, σ^2)

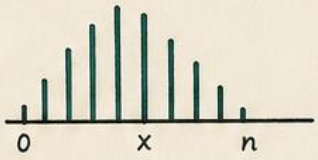
Continuous, symmetric, bell-shaped



Defined by mean (μ) and std dev (σ)

BINOMIAL (n, p)

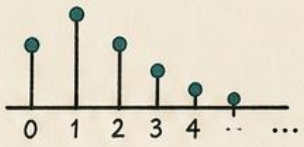
of successes in n independent trials



$X \sim \text{Binomial}(n, p)$
Mean = np
Var = $np(1-p)$

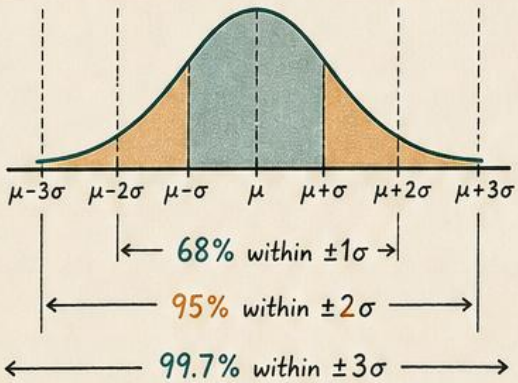
POISSON (λ)

of events in a fixed interval of time/space



$X \sim \text{Poisson}(\lambda)$
Mean = λ
Var = λ

THE 68-95-99.7 RULE (NORMAL)



SAMPLING BIAS — WATCH OUT!

If your sample isn't representative, your results won't be either.

Convenience bias
Easy to reach ≠ representative

Selection bias
Systematic differences distort results

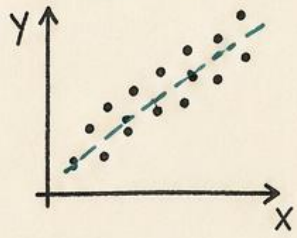
Nonresponse bias
Those who don't respond may differ from those who do

BIAS TODAY, MISLEADING TOMORROW.

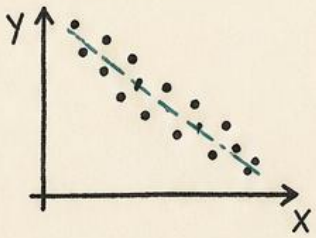
★ *Know the shape before you trust the average.* ★

07 — CORRELATION & RELATIONSHIPS

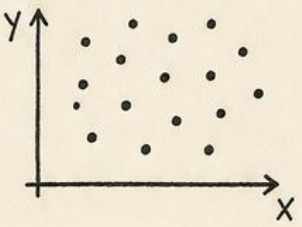
POSITIVE



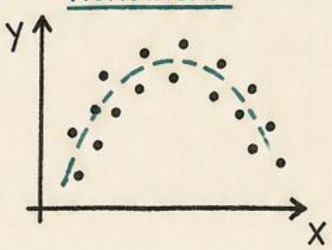
NEGATIVE



NONE (NO CORRELATION)



NONLINEAR



⚠ CORRELATION ≠ CAUSATION

Just because two things move together doesn't mean one causes the other.

PEARSON (r)

- Measures linear relationship
- Assumes continuous & normally distributed data
- Sensitive to outliers
- Range: −1 to +1

Best for linear relationships.

VS

SPEARMAN (ρ)

- Measures monotonic relationship (uses ranks)
- Works with ordinal or non-normal data
- More robust to outliers
- Range: −1 to +1

Best for monotonic (not necessarily linear) relationships.

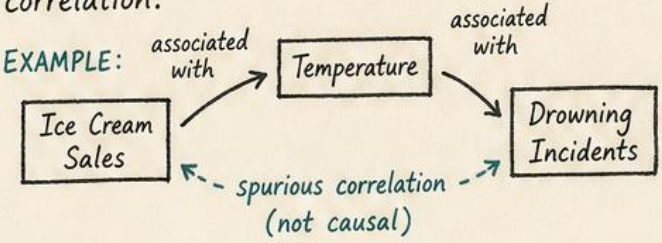
RULE OF THUMB

$r > 0.7$	Strong
$0.3 < r \leq 0.7$	Moderate
$r \leq 0.3$	Weak
$r \approx 0$	No linear relationship

CONFOUNDERS (HIDDEN INFLUENCERS)

A third variable that influences both variables, creating a spurious correlation.

EXAMPLE:



OTHER POSSIBLE CONFOUNDERS:

- ☐ Age
- ☐ Season
- ☐ Socioeconomic status
- ☐ Advertising / Exposure
- ☐ Measurement bias
- ☐ Sampling bias
- ☐ And many more...

★ Correlated ≠ caused. Dig for confounders. ★

08 — SQL For Analysis

1. SELECT & WHERE

Pick and filter the data

```
SELECT col1, col2
FROM table_name
WHERE condition;
```

→ Always filter early for better performance.

2. GROUP BY & HAVING

Aggregate and filter groups

```
SELECT col, COUNT(*)
FROM table_name
GROUP BY col
HAVING COUNT(*) > 10;
```

→ HAVING filters groups, WHERE filters rows.

3. ORDER BY

Sort your results

```
SELECT col1, col2
FROM table_name
ORDER BY col1 DESC;
```

4. Aggregate Functions

Summarize your data

```
COUNT()  – how many
SUM()    – total
AVG()    – average
```

5. JOIN For Analysis

Combine tables for insights

```
SELECT a.col, b.col
FROM table_a a
JOIN table_b b
  ON a.id = b.a_id;
```

→ Know your JOIN types: INNER, LEFT, RIGHT.

6. Window Functions

Calculate across rows

```
SELECT col,
ROW_NUMBER() OVER (ORDER BY col) AS rn,
RANK() OVER (ORDER BY col) AS rnk
FROM table_name;
```

→ Great for rankings, running totals, and de-duplication.

7. CTE (WITH) For Readability

Break logic into steps

```
WITH step1 AS (
  SELECT ...
), step2 AS (
  SELECT ...
)
SELECT * FROM step2;
```

→ Makes complex queries easier to read and debug.

8. CASE WHEN For Buckets

Create categories and labels

```
SELECT col,
CASE
  WHEN col < 10 THEN 'Low'
  WHEN col < 50 THEN 'Medium'
  ELSE 'High'
END AS bucket
FROM table_name;
```

→ Perfect for segmentation!

SQL is how analysts talk to warehouses.

09 — EXCEL / GOOGLE SHEETS

① Pivot Tables



Summarize. Slice. Spot patterns.

Row Labels	Sum of Sales
East	12,450
West	9,320
North	7,850
Grand Total	29,620

Insert → Pivot Table

② VLOOKUP / XLOOKUP

Find. Bring. Done.

ID	Name	Dept
101	Aarav	Sales
102	Diya	Finance
103	Kabir	Ops
...	...	..

Prefer XLOOKUP → handles both directions!

③ FILTER



Return what you need.

=FILTER(Data, Condition)

Product	Region	Sales
Notebook	East	1200
Pen	East	300
Bottle	East	750
...	..	...

④ IF / IFS



Logic > Output

=IF(Condition, Yes, No)

=IFS(Cond1, Out1, Cond2, Out2, ...)

```
=IFS(Sales>=1000,"High",
     Sales>=500,"Med",
     TRUE,"Low")
```

⑤ UNIQUE

Get distinct values.

=UNIQUE(Range)

Mumbai
Delhi
Bengaluru
Chennai
Pune

No more dupes!

⑥ Charts tips



- Pick the right chart.
- Keep it clean.
- Add titles & labels.
- Less is more.



⑦ Data validation

Control input. Reduce errors.

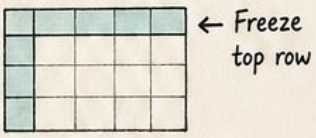


- List • Number • Date
- Custom rules

Data → Data validation

⑧ Freeze panes

Lock what matters. Scroll with ease.



⑨ Power Query / Import idea



Get. Clean. Shape. Load.

- Import from files/DB/web
- Clean & transform
- Automate refresh

Data → Get Data (Power Query)

⑩ Common mistakes



- Hard-coded formulas (use references!)
- No source tab (raw data gets lost)



Keep it dynamic.
Keep it traceable.

Sheets for speed. SQL for scale.

10 — PYTHON PANDAS EDA

① IMPORT

```
import pandas as pd
```

② READ CSV

```
df = pd.read_csv('data.csv')
```

③ HEAD, INFO, DESCRIBE

```
df.head()
df.info()
df.describe(include='all')
```

← first look + shape + summary

④ MISSING VALUES

```
df.isna().sum().sort_values(ascending=False)
# count missing per column
```

⑤ DROP / FILL MISSING

```
df.dropna(subset=['col'], how='any')
df.fillna({'col': 0, 'cat': 'Unknown'})
```

⑥ GROUP BY + AGGREGATE

```
df.groupby('cat').agg({'sales': ['sum','mean']
                       ,'qty': 'count'})
```

⑦ MERGE / JOIN

```
df_merged = df_left.merge(df_right,
            on='id', how='left')
```

⑧ VALUE COUNTS

```
df['cat'].value_counts(dropna=False)
```

← quick freq check

⑨ QUERY / FILTER

```
df.query("sales > 100 & country == 'IN'")
df[df['status'] == 'Active']
```

← clean & readable filters

⑩ TIP

★ Vectorize operations, avoid row loops.

★ pandas = spreadsheet thinking at scale. ★

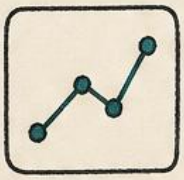
11 — VISUALIZATION RULES

KEY PAGE

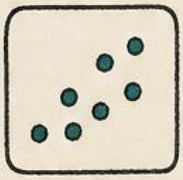
CHART CHOOSER



BAR
for categories



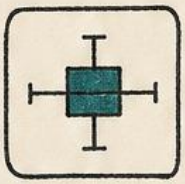
LINE
for trends



SCATTER
for relationships



HISTOGRAM
for distribution



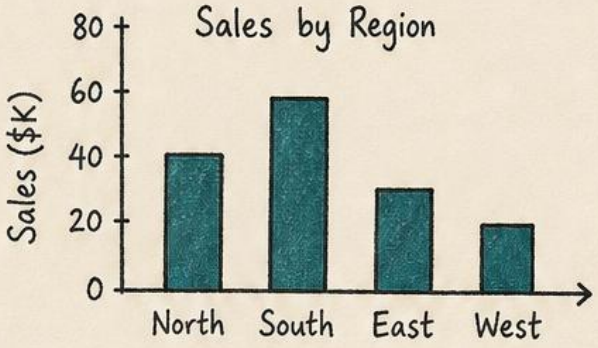
BOX
for spread / outliers



HEATMAP
for matrices

DO

- ✓ Label axes
- ✓ Start bar at zero
- ✓ One message per chart



DON'T

- ✗ 3D pie
- ✗ Dual axis tricks
- ✗ Rainbow overload
- ✗ Chartjunk









★ —— A chart should answer one question. —— ★

12 — METRICS & KPIs

1) METRIC vs KPI

METRIC



A raw measurement.
Answers "what"
Is descriptive.

e.g. Signups, Clicks, Page Views

KPI



A metric tied to a goal.
Answers "so what"
Is decision-driving.

e.g. Activation Rate, Revenue, Retention

2) NORTH STAR METRIC

One metric that best captures the core value you deliver.



Aligns the whole team.
Product, growth, and decisions.

3) RATIO METRICS

Powerful because they normalize scale.

CONVERSION RATE

$$= \frac{\text{Conversions}}{\text{Visitors}}$$

RETENTION RATE

$$= \frac{\text{Users in Period 2}}{\text{Users in Period 1}}$$

Ratios > Raw counts

4) GUARDRAIL METRICS

Keep an eye on what could go wrong while you chase what could go right.



- System Uptime
- Error Rate
- Customer Support Tickets
- Refund Rate

5) VANITY METRICS

Looks good.
Doesn't always mean progress.



- Likes
- Page Views
- Social Followers
- Downloads (without usage)

Don't optimize for applause.
Optimize for impact.

6) DEFINE IT CLEARLY

A metric is only useful if everyone measures it the same way.

1. **NUMERATOR** — Exactly what you are counting.
2. **DENOMINATOR** — The relevant base.
3. **TIME WINDOW** — The exact time period.

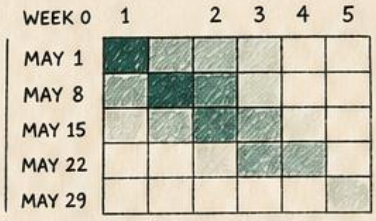
= METRIC

Document the definition.
Share it. Reuse it. Trust it.

7) COHORT THINKING

Group users by shared starting point to see how they behave over time.

Example:
Users who signed up in the same week.



Cohorts reveal patterns that averages hide.

If you can't define it, you can't improve it.

13 — EXPERIMENTS & A/B TESTS

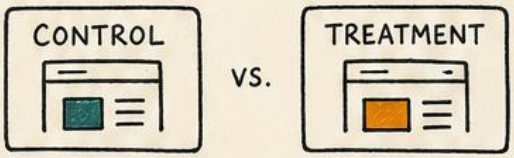
1. HYPOTHESIS



Make a clear, testable prediction.

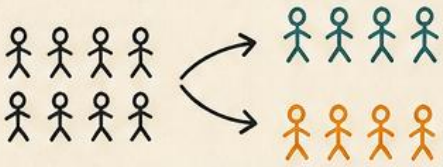
Ex: "New checkout flow will increase conversion."

2. CONTROL vs TREATMENT



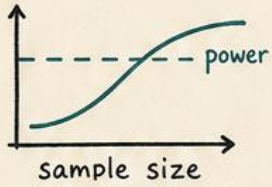
Control = baseline
Treatment = change you test

3. RANDOMIZATION



Randomly assign users to control or treatment to balance known & unknown factors.

4. SAMPLE SIZE INTUITION



Too small = miss real effects.

Too big = waste time/resources.

Aim for enough power (commonly 80%) to detect a meaningful effect.

5. p-VALUE CAUTION (NOT PROOF)



- Low p-value ≠ truth
- High p-value ≠ no effect
- p-value answers: "If no effect, how surprising is our data?"

6. PRACTICAL vs STATISTICAL SIGNIFICANCE



- Statistical: unlikely due to chance
- Practical: big enough to matter

Ask: Will this change decisions or outcomes in the real world?

7. PEEKING PROBLEM



- Checking early inflates false positives.
- Stop looking early, and you'll get fooled.

Let the data speak — at the end.

8. MULTIPLE TESTING

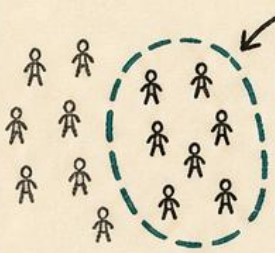


- More tests = more chance of false hits.
- Adjust (e.g., Bonferroni, FDR) or pre-specify key metrics.

☆ Randomize. Pre-register. Don't peek early. ☆

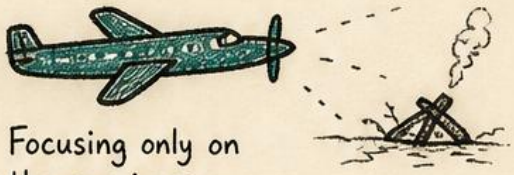
14 – BIAS & PITFALLS

✗ SELECTION BIAS



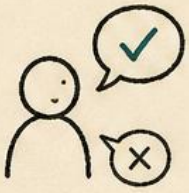
You're only seeing a non-random slice.

✗ SURVIVORSHIP BIAS



Focusing only on the survivors skews the story.

✗ CONFIRMATION BIAS



We seek what confirms our beliefs and ignore the rest.

✗ SIMPSON'S PARADOX

	Group A	Group B	Overall
Treatment	↑	↓	↓
Control	↑	↑	↑

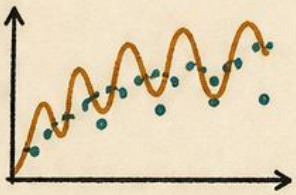
Trends can reverse (or vanish) when groups are combined.

✗ DATA LEAKAGE



Information from the future (or test set) sneaks in. Game over.

✗ OVERFITTING



Model learns the noise, not the signal. Won't generalize.

✗ CHERRY-PICKED TIME RANGES



Zooming in to make everything look great.

✗ SMALL N, LOUD CLAIMS



N = 7

Tiny sample.
Big conclusions.
Bad combo.

☆ Skepticism is a feature, not a bug. ☆

15 – STORYTELLING WITH DATA

1. AUDIENCE FIRST



Know who you're talking to and what they care about.

→ Tailor the story to their world.

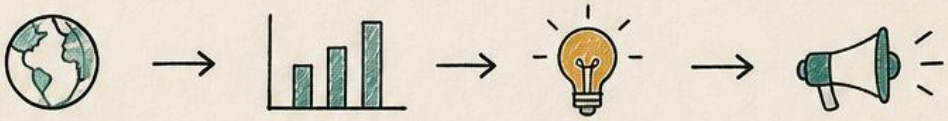
2. ONE INSIGHT PER SLIDE



Focus on one key message.

Keep it clear and uncluttered.

3. CONTEXT → EVIDENCE → SO WHAT → NEXT ACTION



Context
Set the stage.

Evidence
Show the data.

So what
Why it matters.

Next action
What to do.

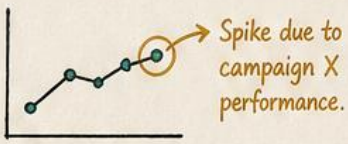
4. TITLE AS CONCLUSION



Sum up the insight in your title.

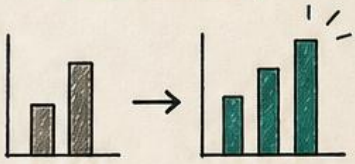
Make it punchy and memorable.

5. ANNOTATION ON CHARTS



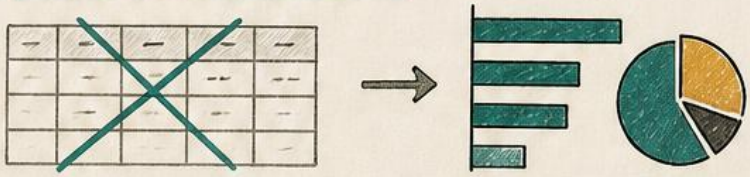
Guide the eye.
Explain the why.

6. BEFORE / AFTER NARRATIVE



Show change.
Highlight impact.

7. AVOID DUMPING TABLES

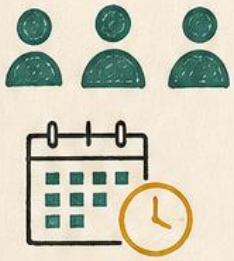


Summarize visually.
Tell the story,
don't dump the data.

★ *Lead with the decision, not the dump.* ★

16 — DASHBOARDS THAT WORK

1. Audience & cadence

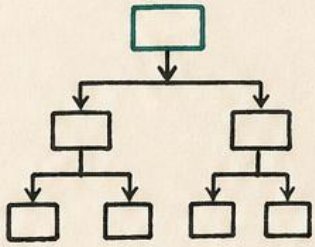


2. Few KPIs



Focus on the vital few.

3. Drill-down path



4. Freshness SLA



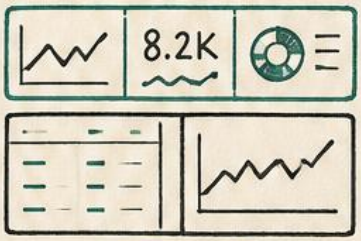
Know how recent is good enough.

5. Alert vs explore

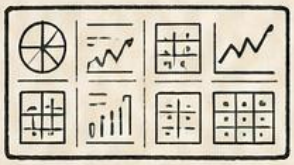
 = Alert

 = Explore

6. Layout: summary top detail below



× ANTI-PATTERNS ×



1. 40 charts



2. no filters



3. vanity vanity vanity

★ A dashboard answers recurring questions fast. ★

17 — ANALYST WORKFLOW

1 BRIEF



Understand problem & objectives

2 SCOPING



Define scope, metrics, deliverables & success criteria

3 DATA ACCESS



Identify sources & get the right access

4 CLEANING



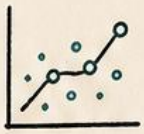
Validate, fix issues, handle missing values

5 EDA



Explore data, find patterns, spot anomalies

6 ANALYSIS



Answer questions, run tests, build models

7 REVIEW



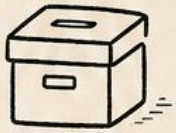
Validate approach & findings with stakeholders

8 SHARE



Communicate insights clearly & actionably

9 ARCHIVE



Save work, document, & close loop

REPRODUCIBILITY



Notebooks
Document steps, explanations, results



SQL Files
Store queries, transformations, & logic



Docs
Record decisions, assumptions, definitions

STAKEHOLDER QUESTIONS

- ☑ What is the business problem?
- ☑ What decision will this inform?
- ☑ Who is the audience?
- ☑ What data defines success?
- ☑ What time period & granularity?
- ☑ Are there constraints or risks?
- ☑ What are the key metrics?
- ☑ How will we validate findings?
- ☑ What are the limitations?
- ☑ What is the next best action?

GOOD HABITS



Start with the right question



Keep it simple & focused



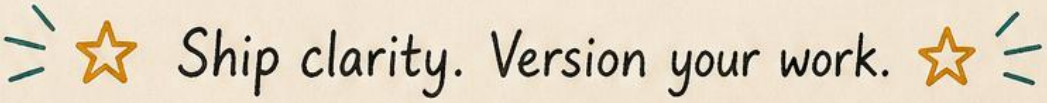
Document as you go



Iterate based on feedback



Mind data quality & privacy



18 – CHEAT SHEET

Copy. Run. Decide.

① CLEANING CHECKLIST

- ☑ Column names clean?
- ☑ Types correct?
- ☑ Missing values handled?
- ☑ Duplicates checked?
- ☑ Outliers reviewed?
- ☑ Ranges & formats valid?
- ☑ Consistent categories?
- ☑ Trim spaces / text case?
- ☑ Dates parsed correctly?
- ☑ Final shape looks right?

② STAT FORMULAS

MEAN

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

MEDIAN

odd n: middle value

even n: $\frac{x_{(n/2)} + x_{(n/2+1)}}{2}$

IQR

$IQR = Q_3 - Q_1$

(Q_1 = 25th pct, Q_3 = 75th pct)

Outliers: $< Q_1 - 1.5 \cdot IQR$ or $> Q_3 + 1.5 \cdot IQR$

③ SQL CORE VERBS

SELECT	– choose columns
FROM	– choose table(s)
WHERE	– filter rows
GROUP BY	– group rows
HAVING	– filter groups
JOIN	– combine tables
ORDER BY	– sort rows
LIMIT	– limit rows
DISTINCT	– unique rows
AS	– alias / rename

④ PANDAS ONE-LINERS

```
df.head()                 # first rows
df.info()                 # overview
df.describe()             # summary stats
df.isna().sum()           # missing count
df.duplicated().sum()     # dup rows
df['col'].nunique()       # unique count
df['col'].value_counts()  # freq table
df[df['col'].isna()]      # missing rows
df.dropna()               # drop missing
df.fillna(0)              # fill missing
df.groupby('col')['x'].agg(['mean','count'])
                          # group summary
df.sort_values('col')     # sort
df[df['x'] > 0]           # filter
df.rename(columns={'old':'new'})  # rename
```

⑤ CHART PICKER (MINI)



Compare categories → Bar / Column



Trend over time → Line



Compare 2+ trends → Line (multi)



Part of whole → Pie / Donut



Relationship → Scatter



Distribution → Histogram



Spread & outliers → Boxplot



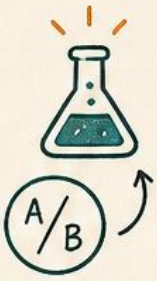
Matrix / correlation → Heatmap

⑥ METRIC DEFINITION TEMPLATE

- Name: _____
- Definition: _____
- Formula / Logic: _____
- Unit: _____
- Higher is better? (Y/N): _____
- Data Source: _____
- Notes / Caveats: _____

⑦ A/B TEST CHECKLIST

- ☐ Define hypothesis & success metric
- ☐ Randomize users (or use holdout)
- ☐ Set sample size / duration
- ☐ Check data quality & tracking
- ☐ Run test
- ☐ Check assumptions (SRM, outliers)
- ☐ Analyze (lift, CI, p-value)
- ☐ Make decision & document learnings
- ☐ Plan next step



★ *Keep this page beside the laptop.* ★

19 — ROADMAP & MASTER RECAP

LEARNING PATH (WEEKS)

1. Foundations & Types
2. Cleaning & EDA
3. Stats & Distributions
4. SQL & Sheets
5. pandas & Viz
6. Metrics & Experiments
7. Storytelling Practice Interview





MUST-KNOW

- ✓ Cleaning
- ✓ Descriptive stats
- ✓ Correlation ≠ causation
- ✓ SQL groupby
- ✓ Chart choice
- ✓ Bias
- ✓ Story structure

COMMON MISTAKES

- ✗ No question
- ✗ Dirty data ignored
- ✗ Wrong chart
- ✗ Vanity KPIs
- ✗ Peeking A/B
- ✗ Overclaiming

CLEAN DATA. CLEAR CHARTS. HONEST INSIGHTS.